

Bring Them Home

AP Physics 1 → Physics C mechanics

Mission Control: four tiers stepping down to a wall of plot boards. The teams are rows, not rooms.

45	10	6.2	8/10	2	10 · 32 · 3
STOPS	CHOICE (22%)	EFFECTIVE FORMATS	EQUATIONS COMPUTED	SELECT-ONLY CONCEPTS	SELECT · CONSTRUCT · OPERATE

READING Where this campaign stands

Strong already: 8 of 10 equations computed, only two select-only concepts, and the best OPERATE count outside Headwater. The two gaps are impulse and centripetal acceleration, and impulse is mentioned across three stops that look arithmetic, so Phase 0 may clear it. The constraint here is the place: Mission Control is one room with no world to drive, so half the fun formats are unavailable on geometry alone — and the two that fit fit exactly.

NOW The mix as it ships

BALLPARK 11	CHOICE 10	SEQUENCE 7	PROTOCOL 7	TRACE 1	FLY 1	ALLOCATE 1	VERIFY 1	CHAIN 1	CONTROL 1	CLOUD 1	STRESS 1
VALUE 1	ATTEST 1										

CURRICULUM What the syllabus is not getting

EQUATION	THE COURSE OWES	WHAT IT CARRIES	MENTIONED AT	STATUS
$J = F\Delta t = \Delta p$		impulse as the change in momentum	7 · SEQUENCE 8 · BALLPARK 10 · PROTOCOL	real gap
$a_c = v^2 / r$		centripetal acceleration in circular motion	6 · CHOICE	real gap

0 of 2 sit on an instrument stop, whose board arithmetic the coverage check cannot yet read — expect Phase 0 to clear those. 2 are unaffected by it.

TIER Concepts the player only ever picks from a list

#	MECHANISM	CONCEPT	STOPS
6	Circular motion and centripetal force		6
27	Fluid pressure and flow		9

Reported, never failed. Select-tier can be exactly right; these are the candidates, not the defects. The syllabus also lists 1 concept no stop touches at all — **Collisions: elastic and inelastic** — which is expected, and is a different conversation from this one.

FUN What this subject and this place can carry

SPOT

STRONG

a FLIGHT or STRUCT stop

A flight rule **is** a standing instruction, written on the board, and it is replaced mid-shift when the vehicle changes state. No other game in the set has a literal, historically accurate home for this format.

SERVES flight rules and the cost of not noticing a change

NEEDS two instructions, at least two changes, and items only one of them wants

HOLD STRONG

NAV or STRUCT

Hold attitude rate inside a closing band while a leaking thruster pushes. The game already carries one FLY, and the argument that made FLY work — bounded commands on undamped dynamics, the brake has to lead — is the same argument one step harder.

SERVES concept 6, circular motion and centripetal force, off the select-only list
NEEDS a do-nothing run must fail; integrate the disturbances with the control untouched

SWEEP STRONG

stop 8 · NAV

"Small burn, large consequence" is a BALLPARK. Burn duration against arrival error is a curve with a minimum, and the point of the stop is that the curve is steep.

SERVES impulse as the change in momentum, made visible
NEEDS the response plotted only where the player looks

TALLY MEDIUM

a THERM or STRUCT stop

Telemetry samples accumulate and the trend firms up. Right subject, and the room is full of plot boards to put it on. Medium because the game's CLOUD stop already argues about width.

SERVES measurement noise against a real trend

TRIAL REJECT

One room, four tiers, no vehicle and nowhere to drive. Rejected on geometry.

BELT REJECT

No sorting line exists in Mission Control and inventing one would be set dressing for a format.

LOB REJECT

A spacecraft game is the last place to teach that a trajectory has to be felt rather than computed. This is the clearest reject in the set.

SLATES Three appetites — the campaign is zero-sum

No conversion adds a stop. Each changes what an existing stop **does** while leaving what it **teaches** alone. Snapshot before, diff after: curriculumDelivery.mjs bring_them_home --snapshot before.json , then --against before.json .

Minimum honest 3 STOPS · ONE SITTING				
6	d2	NAV	CHOICE → HOLD	What holds it in the curve what holds it in the curve, held rather than named
9	d3	THERM	CHOICE → CHAIN	What the pressure difference is for the pressure difference, and the link that governs the flow
8	d3	NAV	BALLPARK → SWEEP	Small burn, large consequence small burn, large consequence — as a curve
CHOICE 10 → 8 (18%) · moves 10·32·3 → 8·32·5 · effective formats 6.2 → 7.3 · 2 new operate stops, 1 of them arcade				

Balanced 6 STOPS · THE RECOMMENDATION

6	d2	NAV	CHOICE → HOLD	What holds it in the curve
9	d3	THERM	CHOICE → CHAIN	What the pressure difference is for
8	d3	NAV	BALLPARK → SWEEP	Small burn, large consequence
10	d4	STRUCT	PROTOCOL → SPOT	Read rotational motion reading rotational motion under a flight rule that is replaced
7	d3	INTEG	SEQUENCE → TALLY	From force to trajectory change force to trajectory change, accumulated
30	d10	NAV	BALLPARK → PROBE	How far it drifts while nobody is steering a fault traced hand over hand along the bus

CHOICE **10 → 8** (18%) · moves **10·32·3 → 8·29·8** · effective formats **6.2 → 8.6** · **5** new operate stops, 2 of them arcade

Ambitious 9 STOPS · ADDS WORLD WORK

6	d2	NAV	CHOICE → HOLD	What holds it in the curve
9	d3	THERM	CHOICE → CHAIN	What the pressure difference is for
8	d3	NAV	BALLPARK → SWEEP	Small burn, large consequence
10	d4	STRUCT	PROTOCOL → SPOT	Read rotational motion
7	d3	INTEG	SEQUENCE → TALLY	From force to trajectory change
30	d10	NAV	BALLPARK → PROBE	How far it drifts while nobody is steering
20	d7	THERM	PROTOCOL → RESIDUAL	Choose the heat-transfer mechanism what the trajectory fit leaves over
35	d12	STRUCT	BALLPARK → TRIGGER	Spring-mass period a go/no-go threshold written before the number moves
40	d14	NAV	CHOICE → DEGENERACY	Real trajectory error or common clock drift? two controls, one telemetry reading

CHOICE **10 → 7** (16%) · moves **10·32·3 → 7·30·8** · effective formats **6.2 → 10.4** · **5** new operate stops, 2 of them arcade

THE ORDER OF WORK

1. Phase 0 — teach the coverage check to read an instrument's own board, and re-baseline curriculum-debt.json. Rewriting a stop that already teaches the thing is the exact failure the check exists to prevent. **2.** Snapshot. **3.** Edit the book, never the generated content — tools/import-book.mjs books/bring-them-home.yml bring_them_home --verify. **4.** npm run traps: every instrument's trap must fire. **5.** npm run check bring_them_home and npm run drive bring_them_home, right answer and wrong — ask what a player who understood nothing would score before believing any pass mark. **6.** --against the snapshot. A changed format is reported; a changed objective fails.

Counts computed from themes/bring_them_home/content/ via engine/dev/curriculumDelivery.mjs; concept and equation lists authored in tools/syllabus.js. Format verdicts and slates are editorial. Run it with THEME=bring_them_home npm run dev.